SET A

Instructions:

1. *Each question has 5 marks.*
2. *Write in a good handwriting.*

Question 1:

Trace the working of Quick Sort for the array: [65,70,75,80,85,60,55,50,45] assuming the first element as pivot.

Question 2:

Given two sorted lists L1 and L2 having number of elements n1 and n2 respectively. The minimum and maximum number of comparisons needed to merge them to get a single sorted list(n1+n2) is. (n1 <= n2)

Question 3:

Write pseudocode of Quick Sort (non recursive).

SET B

Instructions:

1. *Each question has 5 marks.*
2. *Write in a good handwriting.*

Question 1:

Trace the working of Quick Sort for the array: [65,70,75,80,85,60,55,50,45] assuming the first element as pivot.

Question 2:

What is the worst case for quick sort and it's time complexity in worst case?

Question 3:

Write pseudocode of Merge Sort and also write the time complexity of merge sort (non recursive).

SET C

Instructions:

1. *Each question has 5 marks.*
2. *Write in a good handwriting.*

Question 1:

In applying quick sort to an unsorted list if (n/4) the element is selected as pivot then what is the time complexity of quick sort?

Question 2:

Trace the working of Merge Sort for the array: [45, 23, 53, 12, 9, 30, 78, 61, 42, 18].

Question 3:

Write pseudocode of Merge Sort and also write the time complexity of merge sort. (non recursive)